

Legacy standards as frozen solutions to obsolete optimization problems: a framework for technological change and adaptive re-optimization

Abstract

Persistent standards are often treated either as efficient coordination devices or as path-dependent constraints. Both interpretations can be premature when technologies have changed the information, computation, communication, and control conditions under which an incumbent was formed. We develop a Legacy Standard Audit that separates a historical optimum, a current frictionless optimum, and a transition-adjusted optimum. The protocol was applied to a pre-specified universe of 122 candidates in 12 domains and a frozen set of 7 heterogeneous cases. Public inputs were snapshotted with provenance and checksums before confirmatory estimation. Oracle technical benchmarks identified material static mismatch in simulated condition-informed maintenance and reference-crop weather-adaptive irrigation; occupancy-responsive lighting and bounded price-aware load shifting remained small or uncertain after evidentiary and transition constraints were considered. A traffic-control sensitivity was insufficient for redesign, container compatibility could not be quantified, and the repeated-halving paper ratio returned the pre-specified analytic null. Because objectives and denominators differ, effects are not pooled and no population causal claim is made. The results show that technological change can justify re-audit without implying obsolescence: redesign requires case-specific transition-cost, safety, network, and regulatory evidence.

Keywords: standards; path dependence; technological change; adaptive control; switching costs; reproducibility; technology governance

1. Introduction

Standards coordinate expectations, reduce interface uncertainty, and permit complementary investment. Those same benefits can make an incumbent persistent after the problem environment changes. Research on dominant designs and technological trajectories explains how technologies stabilize and later face discontinuities (Utterback and Abernathy, 1975; Dosi, 1982; Abernathy and Clark, 1985; Tushman and Anderson, 1986; Henderson and Clark, 1990).

Research on increasing returns, compatibility, and infrastructure lock-in explains why persistence can remain privately or socially rational (Farrell and Saloner, 1985; Arthur, 1989; Liebowitz and Margolis, 1995; Pierson, 2000; Unruh, 2000). The analytical error is therefore not persistence itself, but inferring either efficiency or inefficiency from persistence without reconstructing the decision problem.

Digital technologies sharpen this problem. Cheap sensing, computation, communication, and adaptive control can relax information constraints that once favored fixed schedules, fixed intervals, or static allocations. Digital innovation also changes architectures and organizing logics rather than merely improving a component (Yoo et al., 2010; Nambisan et al., 2017). Yet real-time information does not erase compatibility, investment, safety, regulation, or collective-action costs. Standards scholarship accordingly shows that dominance depends on installed bases, participation, governance, and the interaction between standardization and innovation (Allen and Sriram, 2000; Suarez, 2004; Wiegmann et al., 2017; van de Kaa et al., 2018; Blind et al., 2022, 2023).

This article contributes a pre-specified measurement and governance protocol. It does not offer a new general theory of lock-in. Instead, it operationalizes a question that spans standards economics, technological transitions, and adaptive systems: when a technology changes the feasible information or control set, how should an incumbent standard be re-audited without selecting only striking examples or forcing a redesign conclusion? The protocol requires separate estimates of static and transition-adjusted regret, preserves null and insufficient-evidence cases, and forbids pooling outcomes with incomparable objectives.

We screened 122 candidates across 12 domains, froze 7 cases before confirmatory estimation, and grounded the positioning in 32 exact-DOI-verified prior works. The empirical cases are demonstrations of a present-day audit, not complete reconstructions of every historical optimum. That distinction is central to the interpretation.

2. Theory and hypotheses

Path dependence can arise through increasing returns and coordination, but historical persistence alone does not establish market failure. Critical work on escape from lock-in and technological trajectories emphasizes that incumbent systems may be difficult to displace even when

alternatives improve (Cowan and Hultén, 1996; Dolfsma and Leydesdorff, 2009; Heinrich, 2014). Sociological and organizational accounts add that standards are enacted through practices and institutions rather than imposed as purely technical rules (Orlikowski, 1992; Timmermans and Epstein, 2010). Exploration and exploitation create a related governance tension: reassessment has option value, but continual change can destroy coordination value (March, 1991).

Transition research frames system change as a co-evolution of technology, institutions, infrastructure, and user practices (Geels, 2002, 2004, 2005). Governance experiments can reveal transition conditions, but their success is contextual (Bos and Brown, 2012). Standards participation and policy further shape whose objectives and constraints enter a redesign process (Choung et al., 2012; Blind and von Laer, 2022). Consumer switching research likewise shows that comparative technical value is only one component of adoption (Kamolsook et al., 2019).

The framework yields five falsifiable expectations. First, fixed rules created under information scarcity should show larger static mismatch where reliable real-time control becomes feasible. Second, adaptive control should improve the specified objective only when information and control gains exceed error and implementation costs. Third, non-technological changes such as demand, prices, demography, and regulation may explain apparent mismatch and must be modeled or named. Fourth, network and switching costs can rationally preserve an incumbent despite positive frictionless regret. Fifth, some inherited standards should remain near-optimal for their specified objective and appear as null controls.

3. The Legacy Standard Audit

Let the incumbent be x_{inc} . The historical optimum x_h^* minimizes loss under the historical feasible set and parameters; the current frictionless optimum x_c^* minimizes current loss without switching costs; and the transition-adjusted optimum x_T^* adds switching, compatibility, and coordination costs. Figure 1 shows the resulting decision sequence.

$$x_h^* = \arg \min(x \in X_h) L_h(x; \theta_h)$$

$$x_c^* = \arg \min(x \in X_c) L_c(x; \theta_c)$$

$$x_T^* = \arg \min(x \in X_c) [L_c(x; \theta_c) + C_{switch}(x_{inc}, x)]$$

$$R_{static} = L_c(x_{inc}; \theta_c) - L_c(x_{c*}; \theta_c)$$

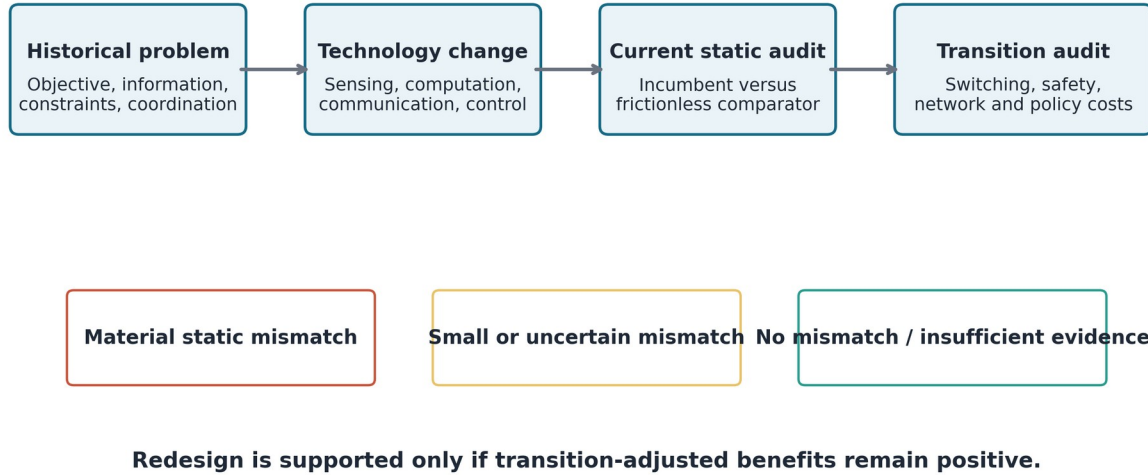


Figure 1. Legacy Standard Audit. Technological change triggers reassessment, but redesign requires transition-adjusted evidence.

Table 1 defines the five audit stages. A case can show positive static regret and still retain the incumbent if discounted benefits do not exceed conversion, coordination, stranded-asset, safety, or regulatory costs. Conversely, a null case is informative because it demonstrates that persistence need not indicate technological inertia.

Table 1. Legacy Standard Audit stages.

Audit stage	Required evidence	Decision output
Historical reconstruction	Objective, information set, feasible set, coordination conditions	Historically defensible incumbent problem
Technology-mediated change	Sensing, computation, communication, automation, adaptive control	Changed parameter or feasible-set claim
Static audit	Incumbent and contemporary comparator under a common objective	Case-specific frictionless regret
Transition audit	Switching, compatibility, safety, regulation, network effects	Transition-adjusted regret or break-even condition
Governance	Frozen cases, uncertainty, null controls, falsification	Material, uncertain, null, or insufficient classification

4. Data and pre-specified methods

Candidate selection preceded confirmatory estimation. The registry covered the built environment, transport, automotive interfaces, work, education, healthcare, packaging, paper and digital legacies, utilities, household and demographic systems, information-scarcity systems, and null controls. Screening used historical evidence, technology relevance, decision clarity, public-data availability, identifiability, counterfactual feasibility, transition-cost feasibility, source quality, and comparability. Expected direction, magnitude, and statistical significance were excluded from selection. The frozen roles were four primary cases, one supplementary case, one network control, and one analytic null control.

Every public input was saved as a local raw snapshot and registered with its URL, identifier, version, access time, retrieval conditions, file size, SHA-256 hash, terms, and completeness. Raw inputs were not overwritten. Analysis configuration, random seed, and bootstrap settings were fixed in a machine-readable configuration. Temporal uncertainty used circular block resampling; the C-MAPSS engine was the independent simulation unit.

Classifications were operational decision labels rather than significance tests. BE03 required the lower uncertainty bound to exceed 20% with no labeled occupied false-off events; WK07 required a positive lower regret bound and relative regret above 5%; UI09 required the lower normalized-regret bound to exceed 5%. UI01 remained uncertain because response and equilibrium were not identified, while TR01 and NC01 required incumbent and transition evidence unavailable in the registered sources. The thresholds organize the frozen audit and are not universal welfare criteria.

BE03 compares a weekday 07:00–19:00 nominal lighting schedule with an oracle occupancy policy using a 10-minute hold. WK07 uses NASA C-MAPSS FD001 simulated run-to-failure trajectories to compare a training-selected fixed replacement age with perfect remaining-life condition information. UI01 conserves daily energy while shifting a bounded fraction of the observed German-Luxembourg load profile toward lower observed day-ahead prices; prices are not re-equilibrated. UI09 calculates FAO-56 reference evapotranspiration from NASA POWER weather and compares a training-period calendar application with perfect-weather weekly requirements. TR01 is limited to a directional demand-allocation proxy. NC01 and NC03 test, respectively, an unquantified compatibility case and an exact geometric null. Source attribution

is UCI Machine Learning Repository (2024) for BE03, National Aeronautics and Space Administration (n.d.) for WK07, Open Power System Data (2020) for UI01, Food and Agriculture Organization of the United Nations (2025) and National Aeronautics and Space Administration POWER Project (n.d.) for UI09, United States Federal Highway Administration (2008) and New York City Department of Transportation (n.d.) for TR01, and International Organization for Standardization (2007, 2020) for the controls.

Table 2 lists the analysis unit, comparator, and primary threat to interpretation for each frozen case. These limitations are part of the result rather than post-estimation caveats.

Table 2. Frozen case design and evidentiary limits.

Case	Role	Persisted input	Analysis unit	Contemporary comparator	Critical limitation
BE03	Primary	UCI office occupancy	20,560 minute-observations	Oracle occupancy control, 10-minute hold	No sensor error, power, commissioning, or installation cost
WK07	Primary	NASA C-MAPSS FD001 simulation	100 evaluation units	Perfect remaining-life condition policy	Simulated trajectories; no observed maintenance program
UI01	Primary	OPSD load and day-ahead prices	725 complete days	1.0% bounded load shift	Fixed prices; no tariff assignment or customer response
UI09	Primary	NASA POWER weather and FAO-56	135 evaluation weeks	Perfect-weather reference-crop requirement	No crop coefficient, soil storage, efficiency, or yield model
TR01	Supplementary	NYC directional traffic counts	18 evaluation days	Demand-proportional directional allocation proxy	No verified intersection or deployed timing plan
NC01	Network control	ISO container-standard metadata	Not estimable	Break-even identity only	No fleet, conversion, or network-cost data

Case	Role	Persisted input	Analysis unit	Contemporary comparator	Critical limitation
NC03	Analytic null	ISO 216 geometric principle	Exact analytic objective	Positive solution $r = \sqrt{2}$	Applies only to repeated-halving similarity

5. Results

BE03 retained 20,560 valid minute-observations after removing two embedded repeated header rows. The oracle occupancy policy reduced nominal on-time by 25.7% (95% block interval 18.0%–34.9%) while both policies had zero occupied false-off observations under the retained labels. The result was classified as small or uncertain because schedule choice strongly changed the estimate and deployment costs and sensor errors were unavailable.

WK07 selected a fixed replacement age of 134 cycles on the training simulation. In the evaluation simulation, the fixed policy cost rate was 0.00746 and the oracle condition rate was 0.00496, yielding static regret 0.00250 normalized cost per cycle (95% interval 0.00229–0.00271) and relative regret 33.5%. This is a material oracle benchmark, not observed fleet savings.

UI01 retained 725 complete days. With 1.0% of each hourly load available for bounded intraday reallocation, the technical procurement saving was EUR 0.072 per MWh of baseline load (95% block interval 0.068–0.076), equal to 0.19% of baseline procurement cost. The small estimate, fixed-price assumption, and absence of customer response support a small-or-uncertain classification.

UI09 used 135 training and 135 evaluation weeks. The calendar application was 42.35 mm/week, compared with an adaptive mean of 43.11 mm/week. Normalized objective regret was 45.7% (95% block interval 41.5%–49.1%). The result remained positive across pre-specified loss weights, but it is a reference-crop and perfect-weather benchmark with omitted agronomic constraints.

TR01 retained 18 evaluation days and produced a 27.2% reduction in a directional delay proxy (95% block interval 22.3%–31.0%). Because the data do not establish an intersection, incumbent timing plan, pedestrian constraints, queues, saturation flows, or safety outcomes, the case

remained insufficient evidence. NC01 also remained unquantified. NC03 returned the exact positive repeated-halving ratio 1.414214 with numerically zero loss for that narrow objective.

Figure 2 reports primary-case benchmarks on separate scales to prevent false comparability. Table 3 reports the corresponding classifications.

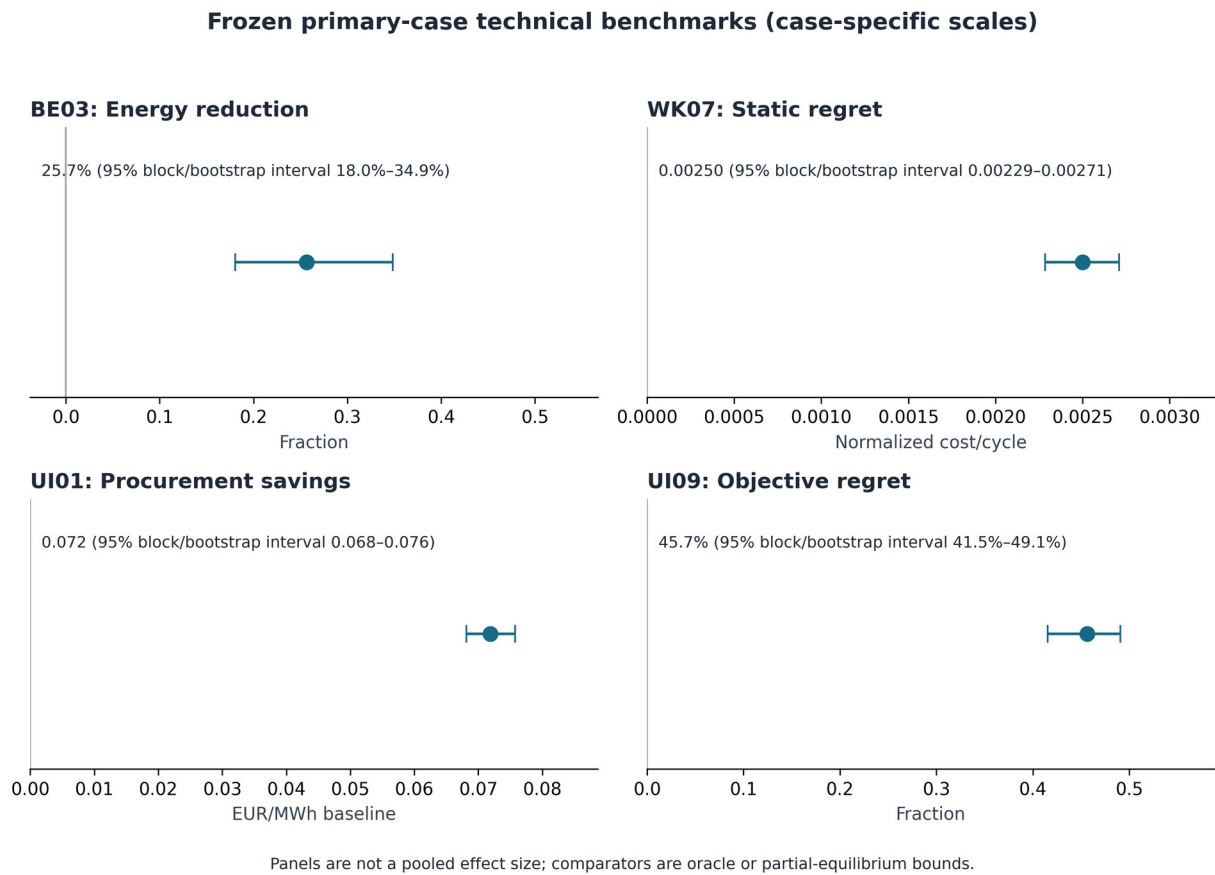


Figure 2. Frozen primary-case technical benchmarks. Each panel uses a case-specific objective and scale; estimates are not pooled.

Table 3. Case-specific technical benchmark results.

Case	Pre-specified metric	Estimate	95% interval	Classification
BE03	Nominal lighting-energy reduction	25.7%	18.0%–34.9%	small or uncertain mismatch
WK07	Static regret, normalized cost/cycle	0.00250	0.00229–0.00271	material mismatch under oracle benchmark
UI01	Procurement saving, EUR/MWh baseline	0.072	0.068–0.076	small or uncertain mismatch

Case	Pre-specified metric	Estimate	95% interval	Classification
UI09	Normalized objective regret	45.7%	41.5%–49.1%	material mismatch under oracle benchmark
TR01	Delay-proxy reduction	27.2%	22.3%–31.0%	insufficient evidence
NC01	Static/transition regret	Not estimated	Not estimated	insufficient evidence
NC03	Repeated-halving loss	4.930e-32	Exact analytic result	no mismatch for repeated halving objective

6. Cross-case synthesis and falsification

Two of 4 primary cases met the pre-specified material-oracle label, while the remaining primary cases were small or uncertain. That count is descriptive. Removing either material case reduced the count to one; excluding simulated C-MAPSS evidence also left one. The cross-case result therefore does not establish a stable population rate of mismatch.

Figure 3 demonstrates why frozen null and insufficient-evidence cases matter. The outcomes span all intended categories rather than converging on a redesign narrative. The strongest evidence is conditional: inexpensive information and control can create measurable static regret under a specified objective, but transition-adjusted recommendations remain unidentified without implementation, network, safety, and coordination costs. Real-time foresight can improve preparedness in dynamic networks (Weber et al., 2015), but it cannot substitute for case-specific institutional evidence.

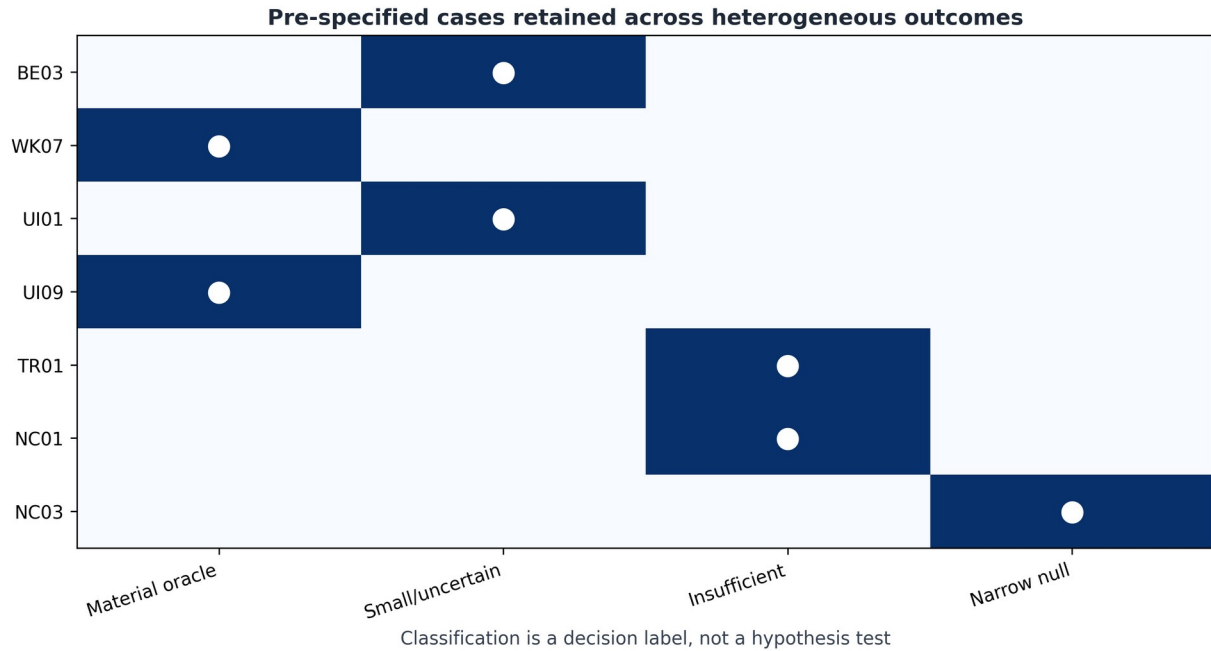


Figure 3. Classifications of all pre-specified frozen cases. Categories are decision labels, not hypothesis-test outcomes.

Table 4. Cross-case falsification and source-quality checks.

Test	Omitted case	Primary retained	Material oracle	Uncertain/ insufficient	Interpretation
baseline primary classification count		4	2	2	Classification count only; incomparable case outcomes are not pooled into a universal effect size.
leave one case and domain out	BE03	3	2	1	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.

Test	Omitted case	Primary retained	Material oracle	Uncertain/ insufficient	Interpretation
leave one case and domain out	WK07	3	1	2	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
leave one case and domain out	UI01	3	2	1	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
leave one case and domain out	UI09	3	1	2	No leave-one-out result licenses a population causal claim; the frozen cases are a designed audit set.
exclude simulated case	WK07	3	1	2	Source-quality sensitivity excluding C-MAPSS simulation.
negative controls		4	2	2	NC03 returns the pre-specified analytic null; NC01 remains unquantified because transition-cost evidence is absent.

7. Discussion

The main contribution is procedural. A standard should be audited as the output of a decision problem, not as an old object. The procedure begins by specifying the historical objective and

constraints, identifies which constraints a technology plausibly changed, estimates a current technical counterfactual, and then asks whether transition costs reverse the ranking. This ordering prevents two symmetric errors: celebrating persistence as evidence of efficiency and treating positive static regret as sufficient evidence for redesign.

The cases illustrate three technology mechanisms. Occupancy detection and weather data reduce the cost of sensing current state. Prognostics and interval metering reduce the cost of computing and communicating condition-dependent actions. Responsive control changes the feasible frequency of adjustment. Yet the same cases show why technological capability is not welfare evidence. Sensor errors, response costs, market equilibrium, crop biology, safety, and installed-base compatibility can dominate a frictionless gain.

For technology governance, the implication is to institutionalize periodic audits rather than automatic sunset rules. The audit trigger should be a measurable change in information, control, production, or coordination conditions. The output should be a transparent classification with an explicit evidence gap. Regulators and standards bodies can then commission the missing transition evidence before altering a coordinated system. Multi-mode standardization and participation research suggests that this process should include affected implementers and complementors rather than treat the technical counterfactual as dispositive (Wiegmann et al., 2017; Blind and von Laer, 2022).

The empirical results also discipline the novelty claim. Existing research already explains lock-in, standards battles, transition governance, digital innovation, and switching behavior. The added value here is a reproducible protocol that freezes case selection, separates static from transition-adjusted regret, retains negative controls, and records when available data are insufficient. This methodological synthesis is narrower than a general theory but more falsifiable than a collection of historical examples.

8. Limitations

First, the analyses do not fully reconstruct the historical optimum for every case. They are frozen demonstrations of the current-audit stages. Second, four comparators are oracle or proxy bounds; none identifies a realized causal deployment effect. Third, transition-adjusted regret is not numerically estimated because compatible implementation-cost data are absent. Fourth, the case

set is purposive and heterogeneous, so neither the frequency nor magnitude of mismatch generalizes to legacy standards as a population. Fifth, uncertainty intervals quantify sampling variation under the retained model but do not capture all structural uncertainty. Sixth, the largest normalized result, UI09, depends on a normative shortfall penalty and omits crop and soil processes.

Future work should prospectively apply the audit to a second frozen set, acquire case-specific switching and implementation costs, and validate technical comparators against deployed controls. A particularly valuable extension would estimate one full transition-adjusted case from incumbent reconstruction through realized implementation.

9. Conclusion

Technological change can make a fixed standard reassessable without making it obsolete. The Legacy Standard Audit converts that distinction into a pre-specified workflow: reconstruct the decision problem, identify the changed technology constraint, estimate static regret, incorporate transition costs, and retain null or insufficient-evidence outcomes. In the frozen demonstrations, some oracle technical benchmarks were material, others were small, one analytic standard remained exactly aligned with its narrow objective, and a high-network compatibility case could not be quantified. The defensible conclusion is not that inherited standards should be replaced. It is that standards governance should make re-audit reproducible, conditional, and explicit about the evidence still required for change.

Data and code availability

All analysis code, frozen selection records, source registry, checksums, and generated values are provided in the accompanying repository. Public raw data are retrieved from registered sources and retained as immutable local snapshots. Copyrighted ISO sample material is not redistributed; its provenance and local checksum are recorded. The one-command build regenerates analyses, figures, tables, manuscript files, supplement, and audits.

Declaration of generative AI and AI-assisted technologies

During preparation, the authors used Devin, an AI software-engineering assistant built by Cognition AI, to support code generation, document assembly, and consistency checks. Human authors are responsible for source verification, analysis choices, numerical results, interpretation, originality, and the final submitted text. The AI system is not an author.

Declarations

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